ELECTRICAL MACHINES · CH3

Magnetic circuits

Cheat sheet · Day 03 · Mon 26 Oct · pack M3

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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MAGNETIC CIRCUITS (Ch3 first half)          mu0 = 4*pi*1e-7 H/m
mmf = NI = Phi * R_total = sum(H*l)          H = B/(mu_r mu0),  Phi = B*A
R = l/(mu_r mu0 A)   gap: mu_r = 1           series add; parallel Ra*Rb/(Ra+Rb)
L = N^2 / R_total     lambda = N*Phi = L*I    "infinite mu"  =>  R_total = R_gap
GAPS: saw cut / 2 poles "g apart" = ONE gap, area A once
      2 C-cores / U-core + armature = TWO gaps: l_g = 2g, force area = 2A
Energy  E = B^2/(2 mu0) * A * l_g (gap)  ; total = 0.5 L I^2 ; iron term has /mu_r
Force   F = B^2/(2 mu0) * A_force         ; hold mass: B = sqrt(2 mu0 m g / A_force)
Back-emf  E_rms = 4.44 N f B_pk A   (NO reluctance in it; B_pk ∝ V/(N f))
   N = V/(4.44 f B_max A) -> round UP; actual B = V/(4.44 N f A)
   V_max = 4.44 N f B_max A ;  new supply: N ∝ V/f ; N2 = N1*V2/V1 (nearest)
DC current fixes mmf (gap changes flux); AC voltage fixes flux (gap changes current)
Saturation ~1.6-2 T: current large and peaky, F∝I^2 and constant L fail
IRON LOSS  P_H ∝ B^2 f (hysteresis),  P_E ∝ B^2 f^2 (eddy; laminate, Si-steel)
   1) split known loss into H and E   2) new B (given, or B ∝ V/f)
   3) P_H2 = P_H1 (B2/B1)^2 (f2/f1),  P_E2 = P_E1 (B2/B1)^2 (f2/f1)^2   4) add
   Fixed f: P ∝ V^2 (hence R_C = V^2/P_FE).  Fixed V, f up: P_H ∝ 1/f, P_E constant
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